

Box Relations

Box definitions

$$\boxed{b} \equiv \boxed{H} \boxed{S^{1/b}} \boxed{H} \boxed{S^b} \boxed{H} \boxed{S^{1/b}} \quad b \neq 0$$

$$\boxed{A_{0b}} \equiv \boxed{b} \quad b \neq 0$$

$$\boxed{A_{ab}} \equiv \boxed{S^{-b/a}} \boxed{H} \boxed{a} \quad a \neq 0$$

$$\boxed{B_{0b}} \equiv \text{circuit with a dot on the top line and a circle with a plus sign on the bottom line, followed by a crossing}$$

$$\boxed{B_{ab}} \equiv \boxed{S^{-b/a}} \boxed{H} \text{circuit with a dot on the top line and a circle with a plus sign on the bottom line, followed by a crossing} \quad a \neq 0$$

$$\boxed{D_{0b}} \equiv \text{circuit with dots on both the top and bottom lines, followed by a crossing}$$

$$\boxed{D_{ab}} \equiv \boxed{S^{-b/a}} \boxed{H} \text{circuit with dots on both the top and bottom lines, followed by a crossing} \quad a \neq 0$$

$$\boxed{E_b} \equiv \boxed{S^{-b}}$$

$A \leftarrow H$: pushing H through an A box

$$\boxed{H} \boxed{A_{0b}} \equiv_s \boxed{A_{b0}} \quad b \neq 0$$

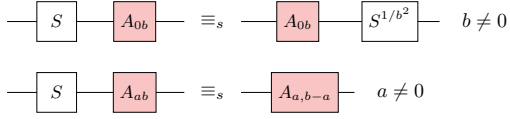
$$\boxed{H} \boxed{A_{a0}} \equiv_s \boxed{A_{0,-a}} \quad a \neq 0$$

$$\boxed{H} \boxed{A_{ab}} \equiv_s \boxed{A_{b,-a}} \boxed{S^{1/(ab)}} \quad a, b \neq 0$$

$$[x]^a \cdot H \equiv_s \text{dir} \cdot [x']^a$$

$A \leftarrow S$: pushing S through an A box

$$[x]^a \cdot S \equiv_s \text{dir} \cdot [x']^a$$



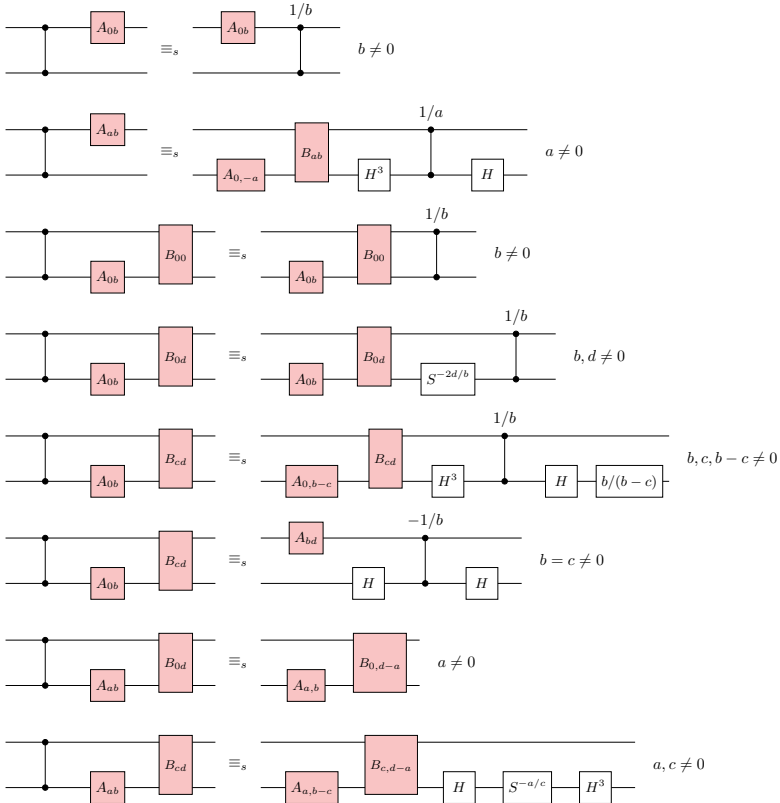
$E \leftarrow S$: pushing S through an E box

$$[b]^e \cdot S \equiv_s [b-1]^e$$



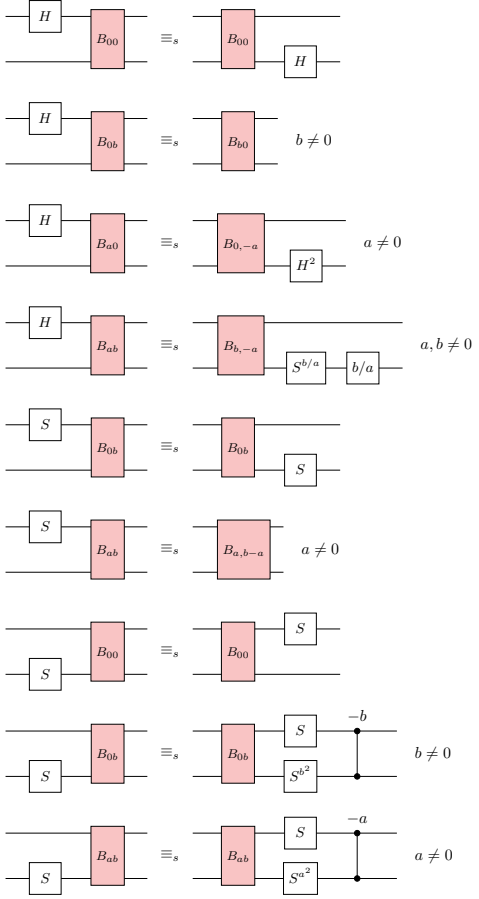
$L \leftarrow CZ$: pushing CZ through an L box

$$[l]^\ell \cdot CZ \equiv_s \text{dir} \cdot [l']^\ell$$



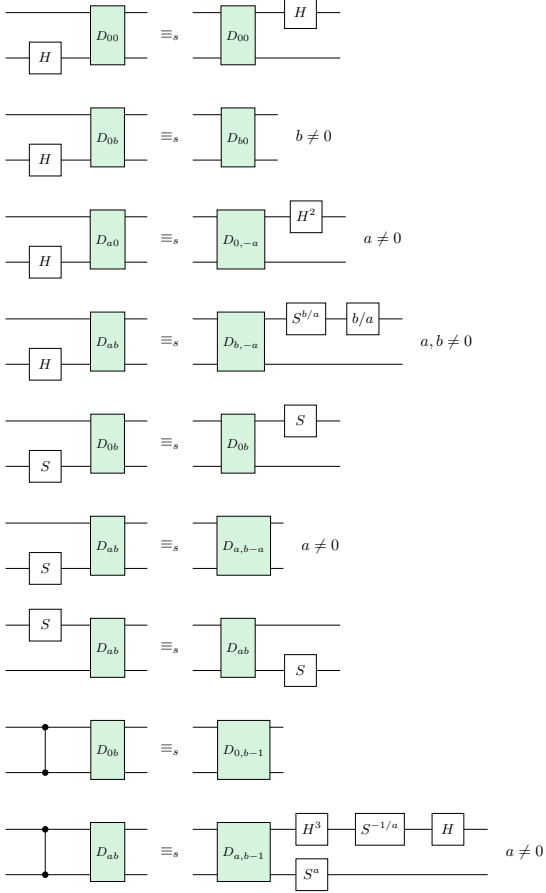
$B \leftarrow H^\uparrow, S^\uparrow, S$: pushing $H^\uparrow, S^\uparrow, S$ through a B box

$$[b]^b \cdot [g]^w \equiv_s \text{dir} \cdot [b']^b$$



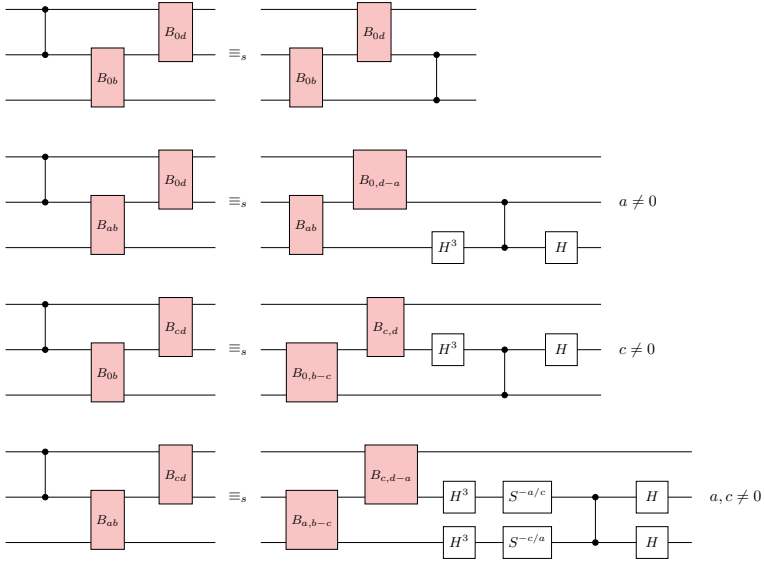
$D \leftarrow H, S^\uparrow, S, CZ$: pushing $H, S^\uparrow, S, CZ$ through a D box

$$[d]^d \cdot [g]^w \equiv_s S^e \cdot \text{dir}^\uparrow \cdot [d']^d$$



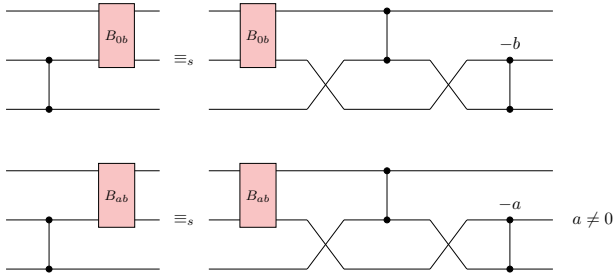
$BB \leftarrow CZ^\uparrow$: pushing $CZ^\uparrow$ through a BB box

$$[vb]^{vb} \cdot CZ^\uparrow \equiv_s \text{dir}\Downarrow \cdot [vb']^{vb}$$



$B^\uparrow \leftarrow CZ$: pushing CZ through a $B^\uparrow$ box

$$[b]^{b\uparrow} \cdot CZ \equiv_s \text{dir} \cdot [b]^{b\uparrow}$$



$DD \leftarrow CZ$: pushing CZ through a DD box

$$[vd]^{vd} \cdot CZ \equiv_s \text{dir}^\uparrow \cdot [vd']^{vd}$$

